

Part 1b: Price Discrimination

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Microeconomics 2 (*Module 1, 2025*).

Suggested readings for this part

Readings in Class Textbooks:

- ▶ **Varian, 8th edition:**
 - Ch 25: Monopoly Behavior
- ▶ **Carlton and Perloff:**
 - Ch 9: Price Discrimination

Overview

- ▶ Up to now, we considered situations in which the monopoly used **single per unit price** (i.e., linear pricing).
- ▶ Recall that marginal revenue is given by

$$MR(q) = p(q) + q \cdot p'(q) \quad \text{where} \quad p'(q) < 0$$

- ▶ To sell more, the firm has to reduce its price — **price discrimination** aims to eliminate this effect.

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- ▶ To sell more, the firm has to reduce its price — **price discrimination** aims to eliminate this effect.

How a monopoly can increase sales without reducing price ?

- ▶ Treat consumers differently — **consumer specific prices**
- ▶ Offer differ. units at differ. prices — **quantity specific prices**
- ▶ **IMPORTANT:** *Price discrimination is feasible for a firm only if resale is banned or too costly.*

Tariffs (Payment Functions)

- ▶ Suppose that consumer i has to pay $T_i(q)$ when he or she buys q units of the good
- ▶ Before we only considered **non-discriminatory linear pricing**. The associated tariff function (payment function) is

$$T_i(q) = p \cdot q \quad \forall i \quad \forall q \geq 0$$

- ▶ In practice, tariff functions around us can be quite different. For example:

$$T^1(q) = \begin{cases} 0 & q = 0 \\ 10 & q > 0 \end{cases} \quad T^2(q) = \begin{cases} 0 & q = 0 \\ 5 + q & q > 0 \end{cases} \quad T^3(q) = \begin{cases} 4q & q \leq 2 \\ 3q & q > 2 \end{cases}$$

- ▶ In fact, consumers can be given a menu of tariffs to choose from

$$\{T^1(q), T^2(q), T^3(q), \dots\}$$

Price Discrimination

We will consider price discrimination in two settings:

1. Unit-Demand Setting
2. General Setting

Price Discrimination

We will consider price discrimination in two settings:

1. Unit-Demand Setting
2. General Setting

And we will cover three types of price discrimination:

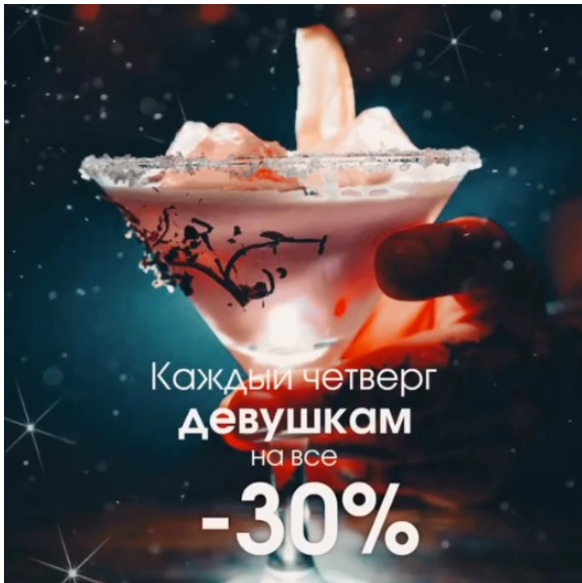
1. First-degree PD
2. Second-degree PD
3. Third-degree PD

Third-degree Price Discrimination

- ▶ linear prices, but **consumer** specific
- ▶ To buy quantity q , consumer i has to pay

$$T_i(q) = p_i \cdot q$$

- ▶ Typically consumer i means a group of consumers with similar observables — location, age, gender, etc
- ▶ Most widespread — senior citizen/student discounts, cars in different countries, etc

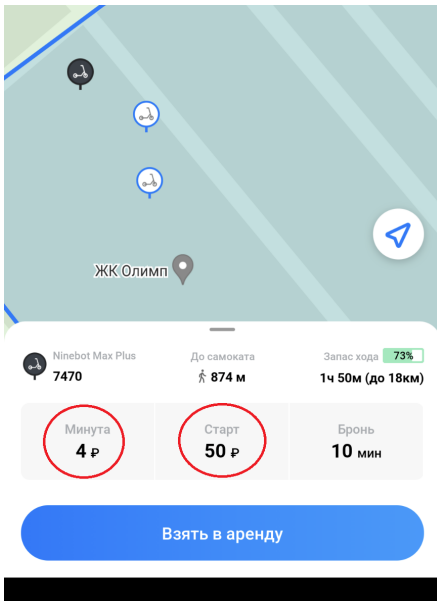


Каждый четверг
девушкам
на все

-30%

Second-degree Price Discrimination

- ▶ only **quantity specific prices**
- ▶ Any tariff functions are allowed as long as they offered to all consumers (no consumer discrimination)
- ▶ Each consumer has to pay $T(q)$ to buy a quantity q .
- ▶ Very widespread — *mobile service, rentals, clothing*.



Rental price in the
Fall 2022:

$$T(q) = \begin{cases} 0 & q = 0 \\ 50 + 4q & q > 0 \end{cases}$$

First-degree (Perfect) Price Discrimination

- ▶ **consumer & quantity** specific prices
- ▶ Consumer i has to pay $T_i(q)$ to buy a quantity q .
- ▶ Often infeasible — needs a lot of consumer data.
- ▶ Often illegal — medical insurance cannot be too discriminatory.
- ▶ Became more feasible in online markets — Amazon, Google, Yandex, Sber store your purchasing data.

Price Discrimination

Unit Demand Setting

What is Unit Demand?

- ▶ Suppose that the good is indivisible
- ▶ A consumer has a **unit demand** whenever there exists $\omega \geq 0$ such that:

$$D(p) = \begin{cases} 1 & 0 \leq p \leq \omega \\ 0 & p > \omega \end{cases}$$

- ▶ This ω is often called **willingness to pay (wtp)** .

Socially Optimal Outcome

- There are N consumers with unit demands such that

$$\omega_1 > \omega_2 > \dots > \omega_N$$

- The monopolist can produce q units of homogeneous good at cost of $C(q)$.
- The socially optimal level of production q^* is given by

$$q^* = \operatorname{argmax}_{k \in \{1, \dots, N\}} \left[\sum_{i=1}^k \omega_i - C(k) \right]$$

How to solve a monopoly problem?

- ▶ Suppose there are N consumers with unit demands such that

$$\omega_1 > \omega_2 > \dots > \omega_N$$

- ▶ The monopolist can produce q units at a cost of $C(q)$.
- ▶ Monopoly's profit maximization can be broken down into two steps:
 1. Suppose that the monopoly wants to sell exactly k units of the good. **How should monopoly price these k units to maximize the revenue?**
 2. Pick the optimal number of units to sell q^m accounting for the production cost.

Linear (non-discriminatory) pricing

- ▶ What non-discriminatory price should a monopolist pick?
- ▶ To sell exactly k units, the linear price p must be

$$\omega_{k+1} \leq p \leq \omega_k$$

- ▶ The best linear price to sell k units is given by

$$p = \omega_k \quad \text{and} \quad \text{Revenue} = k \cdot \omega_k$$

- ▶ The monopoly picks q^m to maximize its profit, i.e.:

$$q^m = \operatorname{argmax}_{k \in \{1, \dots, N\}} [k \cdot \omega_k - C(k)] \quad \text{and} \quad p^m = \omega_{q^m}$$

- ▶ Typically (but not always), the monopolist produces strictly less than the socially optimal quantity, so

$$q^m < q^*$$

1st-degree PD

- ▶ To sell k units, the best **consumer specific prices** are

$$p_i = \begin{cases} \omega_i & i < k \\ \omega_k & i \geq k \end{cases} \quad \text{and} \quad \text{Revenue} = \sum_i^k \omega_i$$

- ▶ The monopolist extracts the CS of **every served consumer!**
- ▶ Note that

$$\sum_{i=1}^k \omega_i > k\omega_k \quad \forall k \geq 2$$

where $k\omega_k$ is the revenue without discrimination

- ▶ The monopoly chooses q^m to maximize its profit, i.e.:

$$q^m = \operatorname{argmax}_{k \in \{1, \dots, N\}} \left[\sum_{i=1}^k \omega_i - C(k) \right]$$

- ▶ Since optimization is identical to the one with socially optimal level, we must have

$$q^m = q^*$$

Class Example - 1

- ▶ There are three unit demand consumers with the following wtps:

$$\omega_1 = 10 \quad \omega_2 = 7 \quad \omega_3 = 6 \quad \omega_4 = 4$$

- ▶ The cost function of the monopoly is $C(q) = q^2$.
- ▶ **Question 1.** Determine the socially optimal level of q .
- ▶ **Question 2.** Find the optimal price for the monopoly when it is limited to linear prices (non-discriminatory).
- ▶ **Question 3.** Find the optimal prices for the monopoly when 1-st degree discrimination is allowed.

2nd-degree PD

- ▶ On the one hand, 2nd-degree PD does not allow **consumer specific prices** — any tariff(s) has to be offered to all customers with no exceptions.
- ▶ On the other hand, the **quantity specific prices** have no discriminatory power in unit demand settings (*they are only useful with multi-unit demands, e.g., volume discounts*)
- ▶ Thus, in the unit demand setting with homogeneous good, **the 2nd-degree PD cannot generate any extra gains relative to linear pricing.**

3rd-degree PD

- ▶ The **quantity specific prices** have no discriminatory power in unit demand settings
- ▶ Then 1st-degree PD must fully rely on the discriminatory power of **consumer specific prices**.
- ▶ The 3rd-degree PD allows **consumer specific per unit price**, i.e.,

$$T_i(q) = p_i \cdot q \quad \forall q \geq 0$$

- ▶ **Thus, in the unit demand setting, the 1st-degree PD and 3rd-degree PD are equally powerful and extract all surplus from consumers.**

Summary for the Unit Demand Setting

Assumptions:

- ▶ A monopoly producing **homogeneous** good
- ▶ Multiple customers with **unit demand** for the good

Results:

- ▶ **Consumer specific prices** have full discrimination power.
 - even 3rd-degree PD extracts all surplus from consumers
 - 3rd-degree PD is equivalent to 1st-degree PD
- ▶ **Quantity specific prices** have zero discrimination power.
 - 2nd-degree PD is fully equivalent to linear pricing
 - The assumption of **homogeneous** good is **critical**. Otherwise, the monopoly can engage in product differentiation (creating varieties), and replace **infeasible consumer-specific** prices with **variety-specific** prices tailored after consumers.

Variety-specific pricing: How does it work?



Two consumers with

$$\omega_1 > \omega_2$$

$$p = \begin{cases} \omega_1 & \text{sell one car} \\ \omega_2 & \text{sell both cars} \end{cases}$$

Variety-specific pricing: How does it work?



Two consumers with

$$\omega_1 > \omega_2$$

$$p = \begin{cases} \omega_1 & \text{sell one car} \\ \omega_2 & \text{sell both cars} \end{cases}$$

Can we do better?

Variety-specific pricing: How does it work?



Depending on consumer preferences for the color, it can be possible to sell both cars at prices

$$p_1 = \omega_1$$

$$p_2 = \omega_2$$

Price Discrimination

General Setting

1st-degree PD: Single Consumer

- ▶ Suppose that there is a single consumer with demand $D(p)$ and the inverse $p(q)$
- ▶ *Note that $D(p)$ does not depend on the budget/wealth of the consumer, i.e., - quasilinear utility.*
- ▶ There are two sources of extra profit for the monopoly: *consumer surplus (CS)* and *deadweight loss (DWL)*.
- ▶ It is easy to show that the monopoly can extract all social welfare in this market with a **simple two-part tariff**

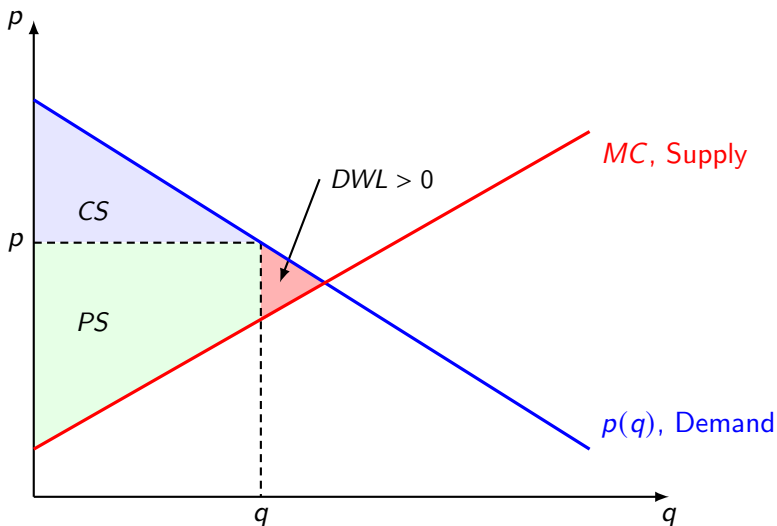
$$T(q) = F + p \cdot q$$

where F is an upfront fee (“entry fee”) and p is per unit price.

- ▶ Let's verify this claim by using a graphical analysis

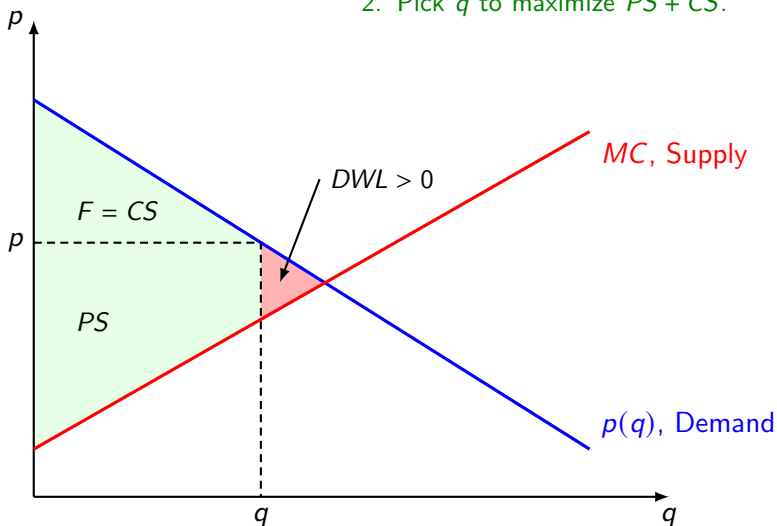
1st degree PD: Graphical Illustration

1. For any q , set flat fee $F = CS$.



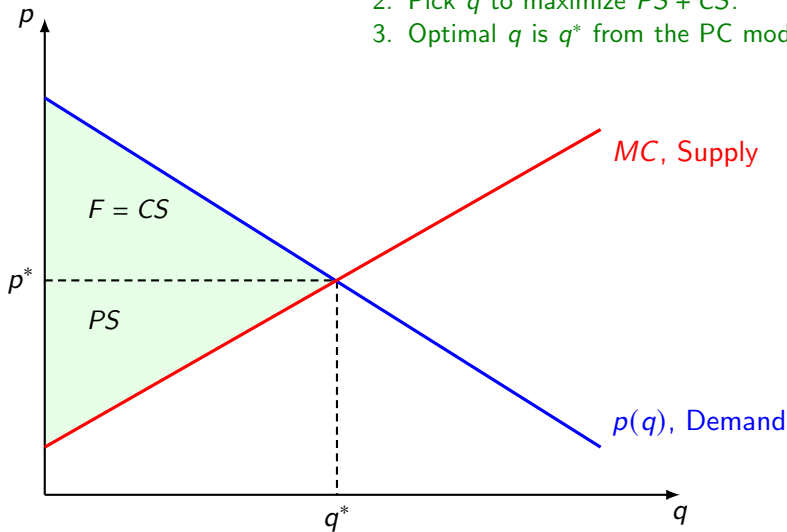
1st degree PD: Graphical Illustration

1. For any q , set flat fee $F = CS$.
2. Pick q to maximize $PS + CS$.



1st degree PD: Graphical Illustration

1. For any q , set flat fee $F = CS$.
2. Pick q to maximize $PS + CS$.
3. Optimal q is q^* from the PC model.



1st degree PD: Single Consumer

- ▶ Recall that for the perfectly competitive outcome, we have

$$q^* : MC(q^*) = p(q^*) \quad \text{and} \quad CS^* = \int_0^{q^*} p(x) dx - p^* \cdot q^*$$

- ▶ Monopoly can extract all SW^* by using the following two-part tariff

$$T(q) = CS^* + p^* \cdot q$$

In other words, charge an “entry fee” of CS^ and then charge per unit price of p^* for any quantity desired.*

- ▶ Given $T(q)$, the consumer is indifferent between buying 0 and q^* (and buying any other quantity is strictly worse).
- ▶ Note that **the social welfare is maximized** ($DWL = 0$), but $CS = 0$ under the 1st-degree PD.

1st degree PD: N Consumers

- ▶ Suppose that there are N consumers with individual demands $D_i(p)$ and inverses $p_i(q)$.
- ▶ Now the monopolist can extract all social welfare by using a consumer specific two-part tariffs (*consumer specific entry fee*)
- ▶ For a perfectly competitive outcome, we have

$$q^* : MC(q^*) = p(q^*) = p^* \quad \text{and} \quad q_i^* = D_i(p^*)$$

- ▶ Monopoly extracts everything from consumer i by using a two-part tariff

$$T_i(q) = F_i + p^* \cdot q \quad \text{where} \quad F_i = CS_i(q_i^*)$$

- ▶ **Summary:** First-degree price discrimination is hard to implement (it needs a lot of data). Also, it is often **illegal** since it is so bad for consumers.

2nd-degree PD: General

- ▶ 1st-degree PD requires **consumer specific prices** (recall entry fees). This approach is infeasible under 2nd-degree PD.
- ▶ The monopolist can use any non-linear tariff(s) that do not depend on identifies of consumers. This is a very large set.

A single non-linear tariff: $T(q) = F + p q + k q^2$

A single two-part tariff: $T(q) = F + p q$

A menu of different tariffs: $\{T_1(q), T_2(q), \dots, T_K(q)\}$

- ▶ Solving for the optimal 2nd-degree PD tariff(s) is a **very challenging problem**. In general, when consumers are different, **the full surplus extraction is infeasible**.
- ▶ In practice, firms tend to use very simple tariff(s): *a couple of two-part tariffs, or a progressive discount on volume, etc ...*

2nd-degree PD: Simplified Setting

Assumptions:

- ▶ Suppose that there are only **two consumers** $i = 1, 2$ with demands $D_i(p)$ and inverses $p_i(q)$
- ▶ Consumer 1 is **high-value** in a sense of

$$D_1(p) > D_2(p) \quad \forall p \in [0, p_1(0))$$

- ▶ Suppose that the monopolist **has to offer the same two-part tariff** to both consumers, i.e.,

$$T(q) = F + p \cdot q$$

What is the best two-part tariff? (*essentially, we just need to solve for two numbers, F and p .*)

2nd-degree PD: Optimal two-part tariff

Given any p , consumer i will buy exactly $q_i = D_i(p)$ as long as F is smaller than consumer's value from the transaction, i.e.,

$$F \leq CS_i(p) .$$

There are only two possibilities here:

1. **Only serve a high-value consumer** — set $F = CS_1 > CS_2$
 - o the monopolist does 1st-degree PD on consumer 1 only
 - o full surplus extraction from consumer 1
2. **Serve both consumers** — set $F = CS_2 < CS_1$
 - o the monopolist maximizes the following profit function

$$\max_p p \cdot D(p) - C(D(p)) + 2 \cdot CS_2(p)$$

- o $D(p)$ is the market demand ($D_1 + D_2$)
- o full surplus extraction from consumer 2
- o partial extraction from consumer 1

2nd-degree PD: Further comments

- ▶ It is better to serve both types when D_2 is relatively close to D_1 . Otherwise, just serve D_1 .
- ▶ Naturally, the monopolist **can do better by going beyond a single two-part tariff** here.
- ▶ For example, a monopolist can offer **a pair of two-part tariffs**:

$$T_1(q) = F_1 + p_1 \cdot q$$

$$T_2(q) = F_2 + p_2 \cdot q$$

such that

$$F_1 > F_2 \quad \text{and} \quad p_1 < p_2$$

and tailor them such that T_1 is more appealing to Consumer 1 and T_2 is more appealing to Consumer 2.

3rd-degree PD: Setting

- ▶ Suppose that there are N consumers with individual demands $D_i(p)$ and inverses $p_i(q)$.
- ▶ Charge every consumer a **different linear price**
- ▶ Consumer i has to pay

$$T_i(q) = p_i \cdot q \quad \forall q \geq 0$$

- ▶ To sell q_i units to Consumer i , the monopoly should set its price to

$$p_i = p_i(q_i)$$

3rd-degree PD: Solution

- **How much to produce?** Find $q_1, \dots, q_N \geq 0$ which solve:

$$\max_{q_1, \dots, q_N \geq 0} \sum_{i=1}^N q_i \cdot p_i(q_i) - C\left(\sum_{i=1}^N q_i\right)$$

- At the optimal solution (**FOC**), we have:

$$MR_i(q_i^*) = MC\left(\sum_{j=1}^N q_j^*\right) \quad \forall i : q_i^* > 0$$

$$MR_i(0) \leq MC\left(\sum_{j=1}^N q_j^*\right) \quad \forall i : q_i^* = 0$$

- For every served consumer, its marginal revenue equals marginal cost of the total production.
- **Hint for solving problems:** *Note that marginal revenues $MR_i(q_i^*)$ are equal to each other for all $q_i^* > 0$.*

Class Example - 2

- ▶ Suppose that there are two consumers with

$$D_1(p) = 10 - p \quad \text{and} \quad D_2(p) = 6 - p$$

- ▶ The cost function of the monopolist is given by

$$C(q) = 0 \quad \forall q \geq 0$$

- ▶ **Question 1:** Solve for a perfectly competitive outcome.
- ▶ **Question 2:** Solve for a monopoly outcome under 1st degree PD.
- ▶ **Question 3:** Solve for a non-discriminatory monopoly outcome.
- ▶ **Question 4:** Solve for a monopoly outcome under 3rd-degree PD.
- ▶ **Question 5:** Solve for a monopoly outcome under 2nd-degree PD (limited to a single two-part tariff).

Summary on Price Discrimination

- ▶ A monopoly always prefers to discriminate (than not). It is impossible for the producer surplus to go down.
- ▶ Unless 1st-degree, other welfare effects are ambiguous.
- ▶ “High” / “Low” demand consumers typically loose/gain, because discrimination $\uparrow/\downarrow$ their prices.
- ▶ The social welfare goes down when the discriminating firm produces less.
- ▶ Or the social welfare goes up (**and even some consumers can gain**) when a discriminating firm produces more.

Some Hints for Solving Problems

Non-discriminatory pricing (same per unit price):

- ▶ **Easy case:** when all $p_i(0)$ are the same, the monopolist always serves all markets. Example of a setting with the same $p_i(0)$

$$D_i(p) = A_i(K - p) \quad \Rightarrow \quad p_i(0) = K$$

- ▶ **Harder case:** when $p_i(0)$ are different across consumers, then it can be optimal for a monopolist to not serve low-value markets.

3rd-degree discriminatory pricing (different per unit price):

- ▶ **Easy case:** A monopolist with a constant marginal costs $C(q) = cq$. Then the monopolist serves all consumers with $p_i(0) > c$. This leads to a very simple system of equations.
- ▶ **Harder case:** Non-constant marginal cost for a monopolist leads to a complex system of equations. Solving requires making “educated guesses” about which consumers are served.

Skill checklist for Part 1b

1. Demand and inverse demand aggregation
2. First-degree PD: consumer specific two-part tariffs (general demand) or individual prices (unit demand)
3. Second-degree PD: optimal tariff in a limited setting (general demand) or linear price (unit demand).
4. Third-degree PD: consumer specific linear prices (general case) or individual prices (unit demand)
5. Compute CS, PS, SW, DWL in each case.